

Sliding Window Seq2seq Modeling for Engagement Estimation

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ABSTRACT

Engagement estimation in human conversations has been one of the most important research issues for natural human-robot interaction. However, previous datasets and studies mainly focus on the video-wise level of engagement estimation, therefore, can hardly reflect human's constantly changing engagement. Fortunately, the Multi-Mediate '23 challenge provides the frame-wise level of engagement estimation task. In this paper, we propose Sliding Window Seq2seq Modeling by BiLSTM and Transformer with powerful sequence modeling capabilities. Our method fully utilizes the global and local multi-modal feature information in the participants' videos and accurately expresses the engagement of the participants at each moment. Our method achieves the state-of-the-art CCC result of 0.71 for engagement estimation on the corresponding test sets.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

engagement estimation; Seq2seq model; sliding window

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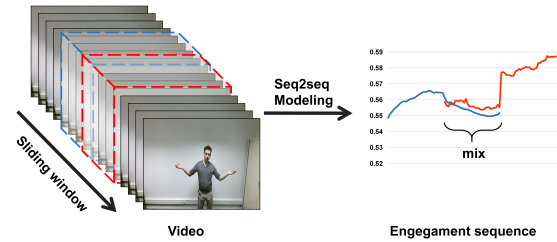


Figure 1: Sliding Window Seq2seq modeling is applied to the training and testing phases, and the overlapping parts need to be mixed (Average fusion) in the testing time.

1 INTRODUCTION

Engagement estimation in conversations is a core task in developing artificial mediators that can understand human behavior and interact with humans. Engagement is closely linked to the previous MultiMediate tasks of eye contact detection [17] as well as backchanneling [8].

The existing problem is generally for the video-wise level of engagement prediction. Most of the existing methods [14, 19] extract features from the whole video sequence or each segment and then process these features by a network based on (Recurrent Neural Networks) RNNs and a fully connected layer to get a tag of engagement. The MultiMediate '23 challenge [16] proposes the frame-wise level of the engagement prediction task. However, the problem arises that the input and output sequences are very long (at least 10,938 or more) and RNNs have difficulty in handling such ultra-long input sequences. Baseline [16] proposes to use a multilayer perceptron network (MLP) for each frame of the input features to output the corresponding engagement, i.e., Point-to-Point (P2P), but this usually leads to the loss of a large amount of video contextual semantic information.

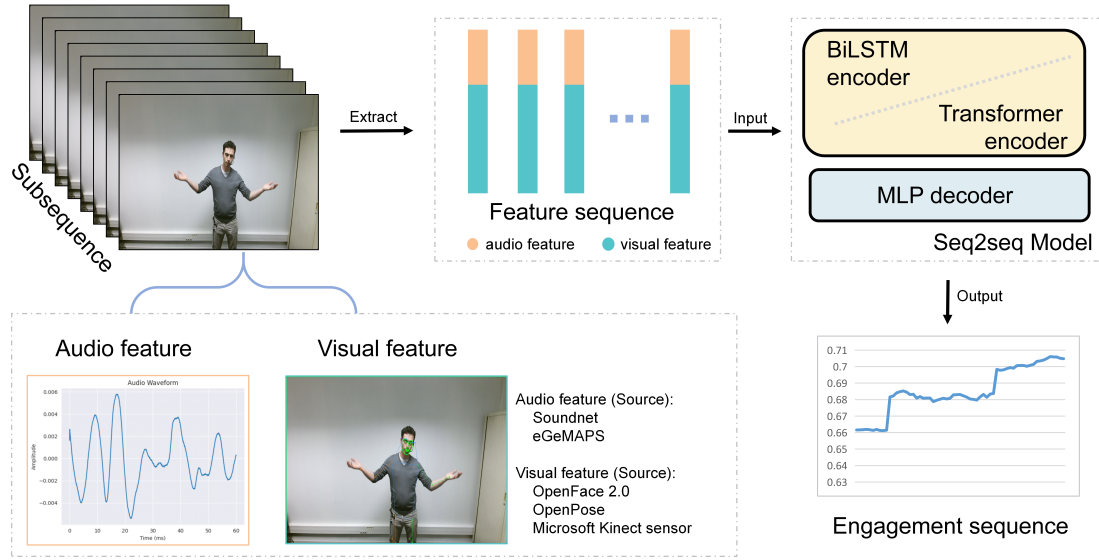


Figure 2: Overview of our method for engagement estimation. After obtaining the video subsequence using "Sliding window", visual frames and speech segments are sampled from the video subsequence and fed into Openface 2.0 and so on to extract multi-modal frame-level feature sequences. The powerful sequence modeling capability of BiLSTM and Transformer is used to output the engagement sequence.

To take full advantage of the contextual information in each participant's video, we model the problem as a sequence-to-sequence (Seq2seq) problem. We design two Seq2seq models based on Bidirectional Long Short-Term Memory Network (BiLSTM) [21] and Transformer [23], to input a sequence of multi-modal features (visual features and audio features) and output the predicted engagement sequence. Since the video is relatively long, we propose to window the input feature sequence, i.e., to slice the ultra-long sequence into multiple subsequences. In order to ensure that the upper and lower subsequences are related, we propose the "Sliding window" method, as shown in Figure 1.

Windowing of the original sequence is guaranteed to take advantage of the information around the current moment (Local information), while "sliding window" connect each subsequence to take advantage of the information of the whole video to estimate the engagement at the current moment (Global information). We validate that our method significantly outperforms the previous P2P baseline at the MultiMediate '23 challenge [16].

2 RELATED WORK

Engagement Estimation. In human-robot dialogue, engagement is an important metric that can be used to assess the quality and effectiveness of human-computer interaction. In recent years, many studies [3, 20] have been conducted to investigate and study engagement estimation from different perspectives. For example, Rich et al. [18] proposed a backchannel-based human-robot interaction engagement recognition module that automatically identifies user engagement. Guhan et al. [10] studied the engagement of mental health patients based on videos of healthy patients.

There are also several more studies that focus on how to detect student engagement [7, 13, 19], which has important implications in the field of education. Kaur et al. [14], in detecting student engagement, used Local Binary Patterns (LBP) [25], Convolutional 3D (C3D) and statistical temporal features to extract facial features from a sequence of cropped student's faces and used a deep neural network to predict the level of engagement. Similarly, Yang et al. [24] proposed another deep multi-instance neural network that has more input features such as upper body pose combined with the LSTM model to estimate student engagement.

Seq2seq Encoder. Seq2Seq Model is a sequence-to-sequence model for converting a sequence to another sequence. In practice, Seq2Seq Encoder can use a variety of models to implement the encoder part. Among them, Recurrent Neural Network (RNN) [12, 21] and Transformer [23] are two commonly used families of Seq2Seq models.

In recent years, Long Short-Term Memory Network (LSTM) [12] have received much attention for their superior performance compared to state-of-the-art RNNs [9]. LSTM can handle long-term dependencies and avoid the gradient vanishing problem of traditional RNNs. One of the extensions of LSTM is BiLSTM [21]. BiLSTM introduces both forward and backward LSTM layers on top of LSTM to process sequence data.

Transformer [23] is a neural network model based on the Self-Attention mechanism for processing sequential data. Transformer achieves good results in sequence modeling tasks by using Multi-Head Self-Attention (MHSA) to capture long-term dependencies in sequential data. The Transformer encoder can capture the semantic information in the input sequence from a global perspective, as it can consider the information at all positions simultaneously.

3 METHODOLOGY

In this section, we will provide a detailed description of our method. Figure 2 shows the pipeline of our method.

3.1 Task Definitions

Given each participant's long video, estimate their engagement in the conversation at each moment, ranging from 0 (no engagement) to 1 (active engagement). The challenge [16] formulates the engagement estimation problem as a continuous regression problem, where each frame of a video is individually labeled.

3.2 Multi-modal features

We divide this subsection into two parts, visual features and audio features, to respectively describe how to extract multi-modal features from participants' videos.

Visual features. Human nonverbal behaviors, such as facial expressions and body postures, can convey a wealth of information about emotions and intentions. We extract features for each video frame, i.e., image. Referring to the official baseline [16], We extract the head features and pose features of the participants using OpenFace 2.0 [2], OpenPose [5], and Microsoft Kinect sensor, respectively.

Audio features. In order to match the frame rate of the video stream, we set a one-second sliding window with a stride of 40ms to extract the audio segments. Referring to the official baseline, we use Geneva Minimum Acoustic Parameter Set (eGeMAPS) [6] and pre-trained Soundnet [1] for extracting audio features. eGeMAPS [6] consists of 88 acoustic parameters that are commonly applied in tasks such as emotion recognition [22], mood recognition, depression, etc. Soundnet [1] is a deep convolutional neural network dedicated to the feature extraction of audio signals. It converts the original audio signal into high-dimensional acoustic embeddings which can be used for tasks such as classification, retrieval, and clustering of audio signals.

3.3 Seq2seq Modeling

We propose to window the input feature sequence, i.e., to slice the ultra-long sequence into multiple subsequences. The Seq2seq problem can be formalized as follows:

$$y_t, y_{t+1}, y_{t+2}, \dots, y_{t+l-1} = f(x_t, x_{t+1}, x_{t+2}, \dots, x_{t+l-1}) \quad (1)$$

where $[x_t, x_{t+1}, x_{t+2}, \dots, x_{t+l-1}]$ represents the multi-modal feature from t to $t+l-1$ moment, $[y_t, y_{t+1}, y_{t+2}, \dots, y_{t+l-1}]$ represents the estimated engagement subsequence, l represents the length of the setup subsequence, and f is the Seq2seq model.

Then, we designed two Seq2seq models based on BiLSTM [21] and Transformer [23] to solve this Seq2seq prediction problem.

BiLSTM Encoder. The BiLSTM [21] architecture is a type of recursive neural network model. Its organizational structure is shown in Figure 3, with a multi-modal feature sequence input, and BiLSTM as the Seq2seq encoder. The BiLSTM layer captures the contextual dependencies of video segments in both the forward direction and backward directions and then outputs the encoding features of each participant at each moment.

Transformer Encoder. The Transformer encoder [23] consists of several blocks of Multi-Head Self-Attention (MHSA) Mechanism

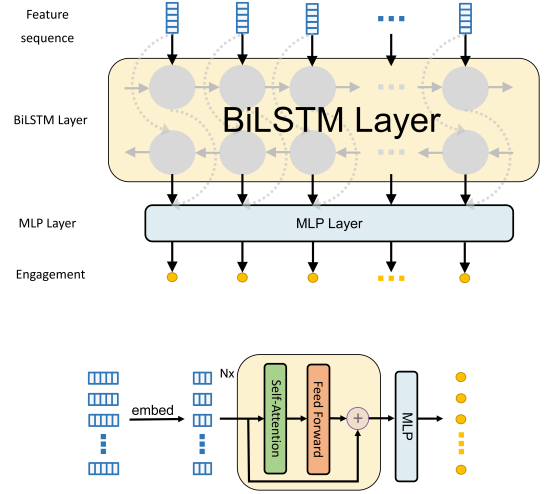


Figure 3: Architectures of BiLSTM and Transformer model.

and Feed Forward Network (FFN). As shown in Figure 3, we first embed the input multi-modal features, then encode them using the Transformer encoder. Similar to BiLSTM, the Transformer encoder encodes the contextual information. However, when processing each input feature, the attention mechanism is able to select important and relevant input features in the sequence based on the semantic information, which leads to a more powerful understanding.

When using a Transformer encoder with a large number of stacked blocks, there can be problems such as network degradation and other issues. To address these problems, we take inspiration from the residual connection method from ResNet [11] and made improvements to the Transformer encoder. By introducing residual connections, the model becomes more stable during training, enhancing the model's generalization ability and helping to avoid overfitting problems. For each block of the transformer encoder, the residual connection can be formalized as:

$$output = \theta \times FFN(MHSA(input)) + (1 - \theta) \times input \quad (2)$$

where θ represents a parameter from 0 to 1 that controls skipping each layer of the connection.

MLP Decoder. We use MLP as the tag decoder of the Seq2seq model. MLP decodes the output of the BiLSTM encoder or Transformer encoder to obtain the estimated engagement. The MLP layer learns the joint relationship between encoding features and outputs the engagement of each participant at each moment.

3.4 Sliding window

From equation (1), the current modeling approach is limited to frames within the current sequence affecting each other, but other sequences of the same video cannot affect the predictions within that sequence. In order to ensure that the upper and lower subsequences are related, we propose the "Sliding window" method, as shown in Figure 1. We set the stride for the "Sliding window" to get the related overlapping subsequences.

"Sliding window" enables global modeling of a single video without consuming additional computational resources. Our modeling of overlapping subsequences using the Seq2seq model is equivalent to modeling the entire video and can be defined as follows:

$$y_0, \dots, y_{L-1} = f(x_0, \dots, x_{L-1}) \quad (3)$$

$$\approx f(x_0, \dots, x_{l-1}) \cup f(x_s, \dots, x_{l+s-1}) \cup \dots \quad (4)$$

where L represents the input sequence length of each video, l represents the subsequence length, s represents stride, $\text{stride} < l$, $[x_0, \dots, x_{L-1}]$ represents complete sequence of multi-modal features for each video, $[x_0, \dots, x_{l-1}]$ represents the 1st multi-modal feature subsequence and $[x_s, \dots, x_{l+s-1}]$ represents the 2nd multi-modal feature subsequence.

We apply the "Sliding window" in both the training and testing process and the overlapping parts need to be mixed equally during testing to get the final result, as shown in Figure 1.

4 EXPERIMENTS

This section describes the dataset used to train and evaluate the engagement estimation, the general experiment setup, metrics, and the results of the experiment.

4.1 Dataset

The challenge [16] used the published NOvix eXpert Interaction (NOXI) database [4]. NOXI includes interactions recorded in three locations (France, Germany and UK), in eight languages (English, French, German, Spanish, Indonesian, Arabic, Dutch, and Italian), discussing a wide range of topics. The dataset provides over 25 hours (x2) of recorded dyadic interactions in a natural environment with synchronized audio, video (25fps) and motion capture data (using Kinect 2.0). This challenge will use a subset of this corpus containing 48 sessions for training and 16 sessions for testing (75/25 split). In particular, in this challenge, each session is continuously annotated, meaning that each video frame has an engagement score from 0 to 1.

4.2 Experimental Setup

We extracted 1704D multi-modal features, which include 1390D visual features and 314D audio features. The visual features have 690D head features and 700D pose features, including 3D facial landmarks, 3D eye landmarks, and body pose estimation. Audio features have 58D acoustic features and 256D high-dimensional acoustic embedding from Soundnet [1].

The input subsequence length of the Seq2seq Model is 64, and the stride used for the "Sliding window" is 32. BiLSTM encoder has 4 layers with hidden layer parameters of 1,024, 512, 256, and 128. The transformer encoder has an embedding of 768D and then 3 blocks with θ of 0.7. The MLP decoder has 1 hidden layer of 128 neurons with SELU activation function and 1 output layer. There are trained using Adam optimizer with $10e^{-3}$ learning rate. The loss was calculated with the mean squared error (MSE) loss function.

4.3 Metrics

The challenge utilizes the Coordination Correlation Coefficient (CCC) [15] to evaluate the predictions y and true label $\hat{y}$ for the test

Method	Model	Val CCC	Test CCC
P2P(baseline [16])	MLP	0.71	0.59
Seq2seq	BiLSTM	0.80	-
Seq2seq	TF	0.79	-
Sliding window Seq2seq	BiLSTM	0.82	0.69
	TF	0.80	-
	BiLSTM+TF	0.83	0.70
Same as above(5-fold)	BiLSTM+TF	-	0.71

Table 1: Validation and test results for engagement estimation. "TF" is denoted as Transformer.

set. The CCC is defined as:

$$\rho_c = \frac{2\rho\sigma_y\sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2} \quad (5)$$

where ρ is the Pearson correlation coefficient, σ_y and $\sigma_{\hat{y}}$ are the standard deviations of y and $\hat{y}$, and μ_y and $\mu_{\hat{y}}$ are the means of y and $\hat{y}$.

4.4 Results

As shown in Table 1, we show the comparison between our approach and baseline [16], which uses MLP for point-to-point (P2P) prediction, while we use the Seq2seq modeling approach. We introduce BiLSTM and Transformer as the encoder part, and we can see that the CCC metrics are improved by almost 10 points in the validation set. In addition, by introducing the "Sliding window", the metric is further improved and the BiLSTM model can obtain with CCC of 0.69 on the test set. The average fusion of the results of 2 models can reach CCC of 0.70. We repartition the dataset and use 5-fold cross-validation to train a total of 5×2 models and fuse the outputs of the 10 models equally, achieving the best CCC of 0.71.

5 CONCLUSIONS

In this paper, we propose a unified and effective method: Sliding Window Seq2seq Modeling for engagement estimation in human conversations. Specifically, we extract multi-modal features from participants' videos, including visual features and audio features. Then, we obtain subsequences by "Sliding window" and use Seq2seq modeling to obtain the predicted engagement sequence. The experimental results on the validation and the unpublished test sets demonstrate that our method has good generalization ability for the challenge. In the future, we hope to start from the data and obtain richer features to help improve the upper bound of our method and achieve better performance.

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